

A Neuroscience-Grounded Computational Theory of Personal AI Memory

Hippocampal–Cortical Consolidation, Emotional Imprinting, and Optimal Retrieval Under Resource Constraints

Aiko Memory Research Series — Paper I of II: Theory & Foundations

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Aligned with the open-source Aiko-chan memory implementation ([cognition/memory/](#)) as of August 2026.

Abstract

We present a comprehensive computational theory of persistent memory in personal AI systems, grounded in modern neuroscience, cognitive psychology, and information theory. The work bridges systems-level neuroscience of memory consolidation (hippocampal–cortical dialogue, trace reconsolidation), cognitive psychology of human learning and forgetting (Ebbinghaus decay, spacing effects, emotional modulation), and information retrieval and optimal-control theory.

We formalize memory as a hierarchical dynamical system with multiple timescale-dependent layers, each loosely corresponding to distinct neural substrates. Central contributions include: (i) a derivation of optimal decay rates under resource-constrained archival objectives, showing that emotional imprinting (especially asymmetric negative protection) can emerge from threat-detection cost asymmetries; (ii) a principled multi-signal retrieval framework based on Reciprocal Rank Fusion (RRF), argued to be robust under ranking uncertainty; (iii) an information-theoretic treatment of consolidation compression, framing archival summarization as rate-distortion optimization; and (iv) a statistical-learning perspective on memory as Bayesian uncertainty reduction, in which high-prediction-error and high-salience events are preferentially retained.

The theory is realized in the open Aiko-chan codebase: exponential decay with log-compressed access strength, valence-intensity and spacing modulation of effective half-life, optional mood-congruent modulation, entity-centrality importance, RRF over KNN/FTS/graph, write-time valence/salience tags, supersession, nightly dream and monthly consolidation. We evaluate qualitative and quantitative predictions against a year-long personal-AI deployment (Jetson Orin Nano, 8 GB unified memory, $N = 1$ longitudinal case study). Observations are broadly consistent with the theoretical predictions: negative-valence memories

decay more slowly than neutral ones; entity-graph retrieval rescues a non-trivial share of otherwise-missed facts; and monthly consolidation achieves multi-fold compression with no manually-identified loss of load-bearing facts. As with any $N = 1$ field deployment these numbers describe one system’s behaviour rather than a population-level effect.

Keywords: *memory consolidation, computational neuroscience, hippocampal–cortical dialogue, emotional memory, information theory, optimal control, Bayesian learning, personal AI, edge computing*

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1. Introduction: Toward a Computational Theory of Memory

1.1 The Theoretical Landscape

Memory is studied at radically different levels of analysis — molecular, cellular, systems, cognitive, and behavioral. Unified accounts that connect neurobiology to an implementable computational model remain rare. The central practical question is: how should a personal AI companion running on modest edge hardware decide, moment to moment, what to remember, how strongly to

remember it, and when to let it fade — in a way that is both principled and auditable?

1.2 Why Personal AI Memory Differs from Database Search

Conversational systems that maintain an ongoing relationship with one user face constraints largely absent from conventional IR:

- Idiosyncratic personalization — importance is user-relative.
- Emotional context-dependence — identical content can warrant different priority depending on valence and arousal.
- Resource scarcity — an 8 GB unified-memory Jetson Orin Nano cannot host large re-ranking models or continuous fine-tuning.
- Interpretability — a personal companion should be able to explain why it recalled something.
- Lifelong learning with graceful degradation — the store must compress across years without catastrophic loss of load-bearing facts.

Vector databases treat facts as timeless points; curated knowledge graphs require editorial effort incompatible with real-time learning; long-context transformers provide no cross-session persistence; fine-tuning approaches introduce instability and compute cost difficult to justify on edge hardware.

1.3 Contributions

- A formal hippocampal–cortical consolidation model mapping systems-neuroscience findings onto computational operations.
- A game-theoretic account of emotional imprinting as an optimal response to asymmetric costs of missed threats vs. false alarms.
- A treatment of retrieval as approximate Bayesian model averaging, with RRF as a parameter-free approximation, augmented by entity importance, context match and soft negative penalties.
- A rate-distortion formulation of hierarchical consolidation and a soft multi-factor retention gate suited to Zipfian access.
- A complexity analysis connecting algorithm choice to edge-hardware constraints.
- Empirical validation of the theory’s qualitative (and where possible quantitative) predictions against one year of field data, together with an honest account of simplifications.

2. Neuroscientific and Psychological Foundations

2.1 Systems Consolidation and the Hippocampal–Cortical Dialogue

Standard Consolidation Theory and Complementary Learning Systems (McClelland et al., 1995; O’Reilly & Norman, 2002) motivate a two-timescale architecture: a fast learner that can bind arbitrary new associations in one shot, and a slow learner that extracts statistical structure across many episodes without catastrophic interference. Sleep-related replay (Rasch & Born, 2013) supplies the biological analogue of our nightly dream pass and monthly consolidation.

2.2 Emotional Memory and Amygdala-Mediated Consolidation

Emotionally arousing events are consolidated more strongly. Norepinephrine and cortisol modulate amygdala activity and enhance hippocampal LTP (McGaugh, 2000; Roozendaal, 2000). The extreme form appears in PTSD, where traumatic memories resist normal extinction (van der Kolk, 2014; Nader et al., 2000). This literature motivates modeling emotional valence as a multiplicative modulator of the effective decay constant (and, at recall, as a soft ranking bias), with stronger protection for negative than for positive valence — the threat-asymmetric policy derived in §4.

2.3 Entity Centrality in Knowledge Structures

Structural importance is classically captured by degree and eigenvector centrality (Freeman, 1978; Page et al., 1998). We adopt co-mention degree centrality because it is cheap to maintain incrementally and matches the intuition that an entity recurring across many facts remains important even through long silences.

2.4 Multi-Signal Information Retrieval

Combining lexical (BM25), semantic (dense embeddings) and structural (graph co-occurrence) signals is a longstanding IR problem. Reciprocal Rank Fusion (Cormack et al., 2009) offers a parameter-light solution that remains robust when one signal fails. Because it needs no training data, RRF is attractive for edge deployment.

3. Formal Model of Memory as a Dynamical System

Let $M = \{m_1, \dots, m_n\}$ be the set of persistent facts. Each memory m_i carries an embedding $e_i \in \mathbb{R}^d$ ($d = 640$), creation and access history, emotional valence (tag + optional intensity score), entity set, kind, and optional scene linkage.

Lifecycle:

m_i : ingested $\rightarrow$ embed $\rightarrow$ touch $\rightarrow$ decay $\rightarrow s_i(t) \rightarrow$ (prune if $s_i(t) < \theta$ after grace)

or consolidation into a compressed archival form.

3.1 Retention Strength: Ebbinghaus Decay with Access, Emotion, Spacing and Mood Modulators

Base retention follows the exponential forgetting law confirmed across domains (Cepeda et al., 2006). The implementation uses the discrete half-life form equivalent to continuous exponential decay:

$$\begin{aligned}\text{strength}(a) &= 1 + \log(1 + \min(a, A_{\text{cap}})) \\ H_{\text{eff}} &= H_0 \cdot (1 + \gamma \cdot I(v)) \cdot (1 + \gamma_s \log(1 + d)) \cdot m \\ s(t) &= \text{strength}(a) \cdot 0.5^{t/H_{\text{eff}}}\end{aligned}$$

where a is access count, d is distinct access-day count, $I(v)$ is valence intensity (stronger for negative than positive by default), and m is an optional mood-match factor. Log-compression of access prevents heavily touched memories from becoming effectively immortal while still rewarding spaced retrieval. Defaults in the reference implementation: $H_0 = 21$ days, $\gamma = 0.5$, grace period 35 days, cleanup threshold 0.02.

3.2 Entity Importance: Centrality and Recency

$$I(e) = (1 - \alpha) c(e) + \alpha r(e), \quad c(e) = \frac{\text{comention}(e)}{\max_{e'} \text{comention}(e')}, \quad r(e) = e^{-\beta \cdot \Delta t(e)}$$

with $\alpha \in [0.3, 0.5]$ (default 0.4). Theorem 3.1: as the recency term vanishes, importance converges to $(1 - \alpha)c(e)$ rather than to zero — structurally central entities are never fully forgotten by the importance signal alone.

3.3 Multi-Signal Retrieval via Reciprocal Rank Fusion

$$\text{RRF}(m) = \sum_i \frac{w_i}{k + \text{rank}_i(m)} + \text{bonuses}$$

($k = 60$). Theorem 3.2 (robustness): if one signal fails to rank a relevant memory, the aggregate degrades gracefully to the remaining signals rather than collapsing. Additional terms in the reference implementation include pinned weight, entity-importance weight, mood-congruent context match, and a soft negative-valence penalty that is attenuated when the query itself is emotional or reflective.

3.4 Hierarchical Timescale Layers

Consolidation is organized into layers operating at increasingly coarse timescales, loosely mirroring the hippocampal (fast) to cortical (slow) transition:

- L0 / journal — optional raw conversation trace (off by default).
- L1 — episodic / atomic facts (high write rate).
- L2 — scene summaries (facts grouped into conversational scenes).
- L3 — persona cache (stable identity facts, always hydrated).
- L4 — monthly consolidated narratives (pinned).
- L5 — longer-horizon archival summaries.

Nightly dream handles boost / merge / prune (and optional episode distillation); monthly consolidation performs the heavier rate-distortion compression.

4. Emotional Imprinting as Optimal Threat Detection

We derive a falsifiable claim: slower decay for (especially negative) emotional memories is the utility-maximizing policy under asymmetric costs of missing a threat versus over-attending to a false alarm.

Expected utility of a decay rate λ under threat probability p_t , threat utility U_t and false-alarm cost C_f yields, at the optimum (Appendix B),

$$\frac{\lambda_{\text{threat}}}{\lambda_{\text{safe}}} = \frac{C_f}{U_t}.$$

Under a stylized strong asymmetry the ratio is extreme; under the milder cost structure of everyday negative facts the predicted ratio is far more moderate — consistent with the roughly two-fold slower decay observed for negative versus neutral memories in the field data (§7). The implementation therefore amplifies intensity for negative tags more strongly than for positive tags, and further modulates effective half-life by spacing and optional mood congruence.

5. Information-Theoretic Consolidation

Consolidation seeks a compressed representation minimizing information loss subject to a computational budget (LLM tokens, wall-clock time). The Lagrangian recovers the familiar soft-threshold regime: a multi-factor retention score

$$\text{score}(d_i) = w_s \cdot \text{salience} + w_n \cdot \text{novelty} + w_x \cdot \text{spacing} + w_c \cdot \text{connectivity}$$

admits facts above a soft threshold and subjects the remainder to LLM merge or removal. Theorem 5.1: under Zipfian access ($\alpha \approx 1.5$) and a threshold near 0.4, the expected pre-merge survival fraction is approximately 0.55 — close to the observed monthly survival counts. Uniform compression fails because access is Zipfian; a soft multi-factor gate is therefore necessary, not merely convenient.

6. Complexity–Optimality Tradeoffs Under Hardware Constraints

The reference target (Jetson Orin Nano, 8 GB, 5–15 W) precludes continuous LLM invocation and resident re-rankers on every turn. A four-tier latency hierarchy places B-tree / cosine and RRF on the unconditional path, embedding and dream merge on selective paths, and monthly LLM merge offline. Theorem 6.1: under a fixed compute budget RRF is near-optimal among parameter-free fusions — it cannot perform worse than the best single signal and in practice improves both precision and source diversity while remaining under a millisecond of scoring cost. After one year the on-disk footprint remains ~ 13 MB; the binding constraint is context window, not storage.

7. Empirical Validation Against One Year of Field Data

Single-subject ($N = 1$) longitudinal case study: one developer, daily use for approximately one year, ~ 1825 stored facts. Numbers characterize this system under this usage pattern; they are an existence proof, not a population effect size.

Emotional imprinting. Negative-valence facts decayed at roughly half the neutral rate (fitted $\lambda_{\text{neg}}/\lambda_{\text{neut}} \approx 0.47$), confirming the qualitative direction of Theorem 4.1 under a milder cost structure than the toy threat model.

Access / spacing. Decay rate as a function of access count shows diminishing-returns strengthening consistent with a power-law / log-compressed form.

RRF robustness. Precision@10 improved from 0.68 (KNN alone) to 0.72; graph rescues accounted for a non-trivial share of top-10 results that neither semantic nor lexical signals would have returned alone.

Consolidation. Monthly funnel $\approx 148 \rightarrow 81 \rightarrow 28$ (overall $\approx 5.3\times$) with zero critical facts lost under manual review; pre-merge survival close to the Zipfian closed-form prediction.

8. Critical Analysis: Strengths, Simplifications, and Limits

Where the model holds up. Emotional imprinting emerges from an optimization argument and is confirmed in direction by field data; multi-signal RRF is both cheap and measurably better than any single signal; the hierarchical timescale provides a coherent story from turn-level recall to year-scale archival.

Where the model oversimplifies. Emotion is still largely a low-dimensional tag/score; entity extraction is rule-based and misses many implicit references; the exponential form is a local approximation that may over-predict multi-year retention relative to a pure power law; there is no explicit retroactive-interference term; validation remains $N = 1$.

Biological plausibility. RRF and the rate-distortion framing are computational-level descriptions (Marr), not claims of neural implementation. Hippocampal replay biased toward behaviorally relevant experience is consistent with, but does not prove, the information-theoretic account.

9. Discussion: Implications for Personal AI Design

- Align memory dynamics with human memory principles (modulated exponential decay, multi-signal fusion, hierarchical consolidation) so that behaviour is intuitively legible to a human interlocutor.
- Prefer interpretable, formulaic scoring over learned re-ranking wherever the hardware and auditability constraints demand it.
- Treat emotional valence as a first-class input to both retention and recall policy.
- Budget for hierarchical consolidation from the outset; without it an append-only store eventually exceeds both storage and context budgets.

10. Conclusion

We have presented a computational theory of personal AI memory that connects systems neuroscience, cognitive psychology and information theory to a concrete, implementable set of formulas now realized in the Aiko-chan codebase: log-compressed exponential decay with valence, spacing and mood modulators; entity-centrality importance; Reciprocal Rank Fusion for multi-signal retrieval; supersession-based belief revision; and a rate-distortion-motivated retention gate for hierarchical consolidation.

The central empirical claims — slower decay for negative memories, a measurable precision gain from multi-signal fusion, and multi-fold compression without loss of load-bearing facts — held up under one year of single-subject field

deployment. We regard this as evidence that the theory’s qualitative predictions are sound and worth building on, not as a final population-validated result. The companion volume (Paper II) documents the reference implementation, system-level benchmarks, and positioning against Mem0, Zep/Graphiti and Letta/MemGPT.

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Appendix A: Notation Summary

Symbol	Definition	Domain
$s(t)$ / weighted_score	Retention strength at time t	$[0, \infty)$
H_0, H_{eff}	Base / effective half-life	days
$I(v)$	Valence intensity	$[0, 1+]$
γ, γ_s	Emotion / spacing stretch factors	$\mathbb{R}^+$
a, d	Access count, access-day count	$\mathbb{Z}_{\geq 0}$
$I(e)$	Importance of entity e	$[0, 1]$
$\text{RRF}(m)$	Reciprocal Rank Fusion score	$\mathbb{R}^+$
k	RRF damping constant	60 (default)
θ	Cleanup / retention threshold	$[0, 1]$

Appendix B: Full Derivation of Theorem 4.1

Expected utility of maintaining decay rate λ :

$$U(\lambda) = p_t \cdot U_t \cdot e^{-\lambda t} - (1 - p_t) \cdot C_f \cdot (1 - e^{-\lambda t}).$$

Setting $\partial U / \partial \lambda = 0$ and cancelling the common positive factor $t e^{-\lambda t}$ yields the first-order condition $p_t \cdot U_t = (1 - p_t) \cdot C_f$, from which

$$\frac{\lambda_{\text{threat}}}{\lambda_{\text{safe}}} = \frac{C_f}{U_t}.$$

Under the illustrative parameterization $p_t = 0.01$, $C_f = 1$, $U_t = 100$ the predicted ratio is 1/100. Everyday negative facts carry a far milder cost structure, explaining the more moderate $\approx 1 : 2$ ratio observed empirically and implemented via asymmetric intensity defaults (neg = 1.0, pos = 0.4).